

Dr. Abdullah Zubair Mohammed

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SUMMARY

Ph.D. in Electrical Engineering (Virginia Tech) with expertise in low-level embedded systems, hardware-assisted security analysis, and physical-layer security, including GPS signal authentication and spoofing detection. Strong background in C-based system programming, FPGA prototyping, and system-level investigation of communication interfaces.

CORE SKILLS

Security and Resilience	Cyber Physical Systems Security, Spoofing, Intention EMI, Fault Injection, Side-Channel Analysis, Secure In-Vehicle Communication
Wireless	Physical Layer: WLAN, BLE, Zigbee, GNSS
Hardware Design	Verilog RTL, Software Defined Radio, FPGA, Circuit Simulation (SPICE)
Simulation & Tools	SPICE, GNURadio, Simulink, Xilinx Vivado, Synopsys DC
Programming	Python, C, Embedded C, Linux Environments, Perl, JavaScript

EDUCATION

2018 - 2024 PhD (Electrical Engineering) at **Virginia Tech**

2014 - 2017 Master of Technology (Communication & Signal Processing) at **IIT Hyderabad, India**

2008 - 2012 Bachelor in Engineering (Electronics & Communication) at **Osmania University, India**

WORK EXPERIENCE

Lead Engineer — Automata Mobility Feb 2026 - Current

Core member of a two-engineer team responsible for end-to-end development of the ADAS/ARAS hardware stack, including system architecture, sensor integration, embedded platforms, vehicle interfaces, and validation. Developing and integrating camera, radar, IMU, edge-compute, and automotive systems.

Engineering Instructor — Loyola University Chicago Aug 2025 - Dec 2025

Teaching faculty in systems engineering with focus on control theory. Courses: Engineering Systems 2, Core Engineering Lab, Engineering Design

Research Associate — Virginia Tech Jan 2025 - Aug 2025

Studied resilience of cyber-physical systems under multi-modal (EMI and Acoustics) injection attacks.

Graduate Research Assistant — Virginia Tech Aug 2018 - Dec 2024

Worked in the area of cyber-physical systems and internet-of-things security, studying potential vulnerabilities, and designing defenses for the physical layer in multiple adversarial environments. My work drew applications from signal processing, embedded systems, electromagnetics, deep learning, RF and analog circuits, and digital circuit design. My research yielded 7 publications in standard security conferences.

Graduate Teaching Assistant — Virginia Tech Aug 2018 - May 2019

For courses Data Structures and Algorithms (using C++), and Intro. to Electrical Engineering.

Design Engineer — Redpine Signals Inc. (now, Silicon Laboratories, Inc) Jul 2017 - Jul 2018

Worked as a base-band engineer towards designing signal processing algorithms for wireless chip-set (WLAN, BLE, and GNSS). Worked on MATLAB at system level followed by Verilog RTL for implementation on FPGAs and ASICs.

Worked on design and implementation (FPGA) of the IEEE 802.5.4 (ZigBee) physical layer algorithms for low-power Internet of Things applications. My research at IITH included 10 publications.

SELECTED PROJECTS

GPS Spoofing Detection via Device Fingerprinting

2019-2023

- My contribution was developing a GPS satellite data acquisition that helps in authenticating satellite transmissions and detects spoofing attacks quickly, cheaply, and robustly.
- Skills used: Software Defined Radio, Signal processing, RF/analog circuits.

Intentional EMI attacks on sensors and actuators in cyber-physical systems (CPS)

2019-2023

- Generated EMI using wide-band attack signals to cause *bit-flips* and manipulate sensor and actuator serial communication (UART, I²C), on wires and PCB traces.
- Skills used: Signal processing, RF/analog circuits, Embedded systems, Electromagnetic theory.

Physical layer data manipulation on CAN bus

2022-2024

- Exploited transmission line nature of the CAN bus in a vehicle to generate transients that can cause controlled bidirectional *bit-flips*, challenging CAN's fundamental physical layer assumptions and impacting the vehicle's safety.
- Skills used: Embedded systems, Transmission line theory, Circuit theory, Vehicular communication network.

Vehicle Fingerprinting

2021-2023

- Identification of vehicle's make, model, and, year based on its response to loop detectors, with an accuracy of 93% using a convolutional neural network.
- Skills used: Signal processing, Software-defined radio, Applied deep learning.

SELECTED PUBLICATIONS

[Google Scholar](#)

Mohammed, Abdullah Zubair and Ryan Gerdes (2025). "From Transients to Flips: Hardware-level Bit Manipulation of In-Vehicle Serial Communication". In: *ACM AsiaCCS 2025*.

Mohammed, Abdullah Zubair, Louis Jenkins, et al. (2024). "The IEMI Effect: On the Efficacy of PCB-level Countermeasures in Adversarial Environments". In: *2024 IEEE Euro Security & Privacy*.

Mohammed, Abdullah Z et al. (2022). "Towards Wireless Spiking of Smart Locks". In: *2022 IEEE Security and Privacy Workshops (SPW)*. IEEE, pp. 251–257.

Mohammed, Abdullah Zubair, Ajay Kumar Nain, et al. (2018). "A residual phase noise compensation method for IEEE 802.15. 4 compliant dual-mode receiver for diverse low power IoT applications". In: *IEEE Internet of Things Journal*.

Dayanıklı, Gökçen Yılmaz, **Mohammed, Abdullah Zubair**, Ryan Gerdes, and Mani Mina (2022). "Wireless manipulation of serial communication". In: *Proceedings of the 2022 ACM on Asia Conference on Computer and Communications Security*.